

B.Tech. Degree IV Semester Examination April 2013

CS/EC/EB/EI 402 MICROPROCESSORS (2006 Scheme)

Time : 3 Hours

Maximum Marks : 100

PART A (Answer ALL questions)

(8 × 5 = 40)

- I. (a) Explain the register array of 8085 microprocessor.
(b) Distinguish between memory mapped I/O and peripheral mapped I/O.
(c) Write the usage of 8085 pins.
 (i) HOLD
 (ii) SID
(d) Write a short note on the various rotate instructions and their functionalities in 8085.
(e) Write short notes on:
 (i) T State
 (ii) Machine cycle
(f) Draw the timing diagram of the instruction MOV A, B.
(g) Write a short note on the various ports available in 8255.
(h) Explain any one operating mode of 8254 programmable timer interface.

PART B

(4 × 15 = 60)

- II. Explain internal architecture of 8085 microprocessor with relevant block diagram. (15)
OR
III. (a) Describe the concept of databus multiplexing and demultiplexing in 8085 with the help of figures. (10)
(b) Explain DMA features. (5)
IV. (a) Explain the various addressing modes available in 8085 with the help of examples. (10)
(b) Explain the sequence of program execution when a subroutine is called and executed. (5)
OR
V. With the help of a flowchart and necessary comments write an 8085 assembly language program to find the largest and smallest of a block of 10 datas. (15)
VI. (a) Draw and explain the timing diagram of LDA 8000H. (10)
(b) Distinguish between maskable and nonmaskable interrupts. (5)
OR
VII. Describe the various types of interrupts and their organisation in 8085. (15)
VIII. (a) Draw the architecture and explain the operation of peripheral chip 8279. (10)
(b) Explain how 8255 I/O ports can be set up for BSR mode of operation. (5)
OR
IX. (a) Write a short note on the concept of memory interfacing. (10)
(b) Draw the architecture and register organisation of peripheral chip 8251. (5)
